SMALL STEPS. BIG FUTURES.

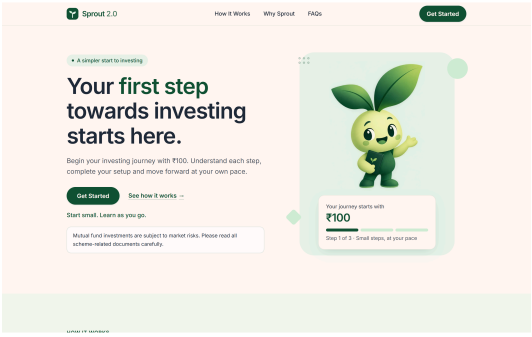
Sprout 2.0 — Beginner-Friendly Investing Starting from ₹100

A complete UI/UX product redesign, design system, responsive web application, and 5-step bilingual (English & हिंदी) onboarding prototype addressing the **77.50% pre-KYC** and **66.67% post-KYC** drop-off in first-time retail investor acquisition.

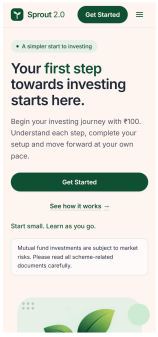
40,000 DOWNLOADS BASELINE · 9,000 KYC (22.50%) · 3,000 FIRST INVESTMENTS (7.50%) · WCAG 2.1 AA VERIFIED · REACT 19 + TS + TAILWIND 4



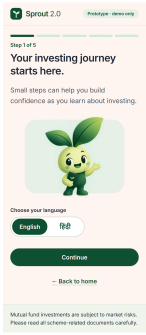
Consistent Geometric Sprout Mascot
Homepage Hero · Step 1 Welcome · Step 5 Pending



Live Desktop Homepage (1440 × 900 Fold) — `src/pages/Home.tsx`



Mobile Homepage (390 × 844)



Step 1: Welcome (390 × 844)

Assignment Deliverables & Traceability Matrix

Deliverable	Scope & Evidence	Repository Path / User-Provided Figma Link
1. Responsive Web App	Desktop (1440px) + Mobile (360px–430px) Homepage & 5-Step Onboarding (0 build/lint errors)	https://github.com/Vaishnavi-khatri2005/sprout-2.0-newtonite (`src/`)
2. Figma Design File	Editable frames, design system tokens, mascot & SVG component assets (user-provided link)	https://www.figma.com/design/67FVguQmfCpgOL2XGjKZge/Sprout-2.0-%E2%80%94-Newtonite-UI-UX-Assignment
3. Figma Prototype	Interactive click-through onboarding flow across English & Hindi frames (user-provided link)	https://www.figma.com/proto/67FVguQmfCpgOL2XGjKZge/Sprout-2.0-%E2%80%94-Newtonite-UI-UX-Assignment?node-id=0-1&t=5Vwakxuj7i3lecnq-1
4. Case Study PDF (in Repo)	10-page Case Study PDF, Markdown source & Figma handoff kit	https://github.com/Vaishnavi-khatri2005/sprout-2.0-newtonite/blob/main/docs/sprout-2.0-case-study.pdf (and local copy at `docs/sprout-2.0-case-study.pdf`)

CONTEXT & PROBLEM FRAMING

Why First-Time Investors in India Hesitate Before & During Onboarding

Sprout is an investing product designed for first-time retail investors in India with an accessible starting amount of **₹100**. Despite strong top-of-funnel acquisition (**40,000 downloads**), the existing product converts only **9,000 users through KYC (22.50%)** and only **3,000 users to a first investment (7.50% of downloads)**. Beginners face three compounding barriers across the landing and onboarding journey:

BARRIER 01 · CAPITAL THRESHOLD

"Investing requires large sums"

First-time earners and students often assume mutual funds require thousands of rupees or deep market expertise. If the **₹100 starting point** is buried below generic copy, hesitant visitors bounce before starting onboarding.

BARRIER 02 · UNPLANNED KYC

Surprise document requirements

Indian regulatory KYC requires a **PAN card**, an **Aadhaar card**, and an **Aadhaar-linked mobile number** for OTPs. Discovering these requirements mid-form forces users to exit the app to search for physical cards.

BARRIER 03 · TONE & LANGUAGE

Intimidating UI & English-only walls

Dense financial jargon, complex trading dashboards, unformatted PAN inputs, and English-only onboarding alienate beginners who need calm, plain-language guidance and regional language support (हिंदी).

Five Core Project Objectives for Sprout 2.0

1. Anchor Value Around "Small Steps" (₹100)

Lead the homepage hero with the **₹100 starting threshold** and the *"Small Steps. Big Futures."* positioning so beginners immediately understand they can learn at a comfortable pace.

2. Surface Document Readiness Before Form Entry

Introduce a transparent 3-item checklist (**PAN card**, **Aadhaar card**, **Mobile number linked to Aadhaar**) on both the homepage and Onboarding Step 2 before asking for any form input.

3. Progressive 5-Step Onboarding Architecture

Decompose onboarding into five single-purpose mobile screens (Welcome → Get Ready → Mobile Verification → KYC Details → Verification Pending) with clear progress (`Step X of 5`) and back navigation.

4. Bilingual Accessibility (English & हिंदी Prototype)

Place a segmented **English / हिंदी** selector on Onboarding Screen 1, dynamically loading `Noto Sans Devanagari` while clearly labeling the Hindi translation as a prototype.

5. Uncompromising Regulatory & Prototype Honesty

Keep the mandatory market-risk disclosure (*"Mutual fund investments are subject to market risks. Please read all scheme-related documents carefully."*) visible above the fold on the homepage and in the footer of every screen. Make all prototype boundaries explicit: never send real SMS, never store or transmit PAN/Aadhaar data, and never promise guaranteed investment returns.

Mapping Business & User Goals to Implemented Product Surfaces

User Question Upon Landing	Sprout 2.0 Design Response	Implemented Component
"How much money do I need to begin?"	Hero copy & illustrative card state: **start with ₹100**; FAQ #1 answers *"Can I start with ₹100?"* opened by default.	`Hero.tsx` , `WhySprout.tsx` , `Faq.tsx`
"What documents will I be asked for?"	Homepage *Getting Ready* section & interactive Step 2 checklist list PAN, Aadhaar, and linked mobile upfront.	`Checklist.tsx` , `Step2Ready.tsx`
"Why do you need my PAN and Aadhaar?"	Step 4 explains in plain language that financial regulations require investor identification, plus format hint `ABCDE1234F` .	`Step4Kyc.tsx` , `Faq.tsx`
"Is my money guaranteed to grow?"	Explicitly states investments are **not risk-free**; market-risk disclaimer appears in Hero, FAQ #3, Footer, and Onboarding footer.	`constants.ts` , `Hero.tsx` , `OnboardingLayout.tsx`

QUANTITATIVE FUNNEL ANALYSIS

Diagnosing the 40,000 → 9,000 → 3,000 Acquisition & Activation Funnel

Using the three baseline figures supplied in the assignment brief (**40,000 downloads**, **9,000 KYC completions**, and **3,000 first investments**), we compute both **stage-to-stage** and **end-to-end (cumulative)** conversion rates, explicitly defining each denominator so there is zero ambiguity between relative and total drop-off.

STAGE 1 · TOP OF FUNNEL

40,000

Total App Downloads (100.00%)

Baseline denominator for cumulative funnel metrics. Represents users with initial curiosity to explore investing.

STAGE 2 · IDENTITY VERIFICATION

9,000 (22.50%)

KYC Completions (9,000 ÷ 40,000)

31,000 users (77.50% of downloads) drop off before completing KYC verification.

STAGE 3 · ACTIVATION

3,000 (7.50% / 33.33%)

First Investments Completed

33.33% of KYC completions (3,000 ÷ 9,000) and **7.50%** of total downloads (3,000 ÷ 40,000).

Exact Conversion & Attrition Table (With Denominator Definitions)

Transition / Metric Name	Exact Formula (Num ÷ Denom)	What the Denominator Represents	Conversion Rate	Drop-Off Count & Percentage
Download → KYC Completion Rate	`9,000 ÷ 40,000`	All **40,000** users who downloaded the app	**22.50%**	**31,000 users lost** (**77.50%** of all downloads fail to finish KYC)
KYC → First Investment Rate *(Stage-to-Stage Conditional Rate)*	`3,000 ÷ 9,000`	Only the **9,000** users who completed KYC verification	**33.33%** (⅓)	**6,000 users lost** (**66.67%** of KYC-verified users never invest; represents **15.00%** of total downloads)
Download → First Investment Rate *(End-to-End Cumulative Rate)*	`3,000 ÷ 40,000`	All **40,000** users who downloaded the app	**7.50%**	**37,000 users lost** (**92.50%** total attrition from download to first investment)

Visual Funnel Proportion & Two-Zone Drop-Off Diagnosis

1. App Downloads (Baseline Cohort) — **40,000 users (100.00% of downloads)**

2. KYC Completed (22.50% of Downloads) — **Leak Zone A: −31,000 users (−77.50%)** — **9,000 users (22.50% of downloads)**

3. First Investment (33.33% of KYC | 7.50% of Downloads) — **Leak Zone B: −6,000 users (−66.67% of KYC)** — **3,000 users (7.50% of downloads)**

DIAGNOSIS · LEAK ZONE A (31,000 PRE-KYC DROP-OFFS)

Why 77.50% abandon before KYC completion

- **Document unreadiness (Hypothesis):** Users start registration without their PAN card or Aadhaar-linked phone nearby and abandon when prompted mid-flow.
- **Input & trust friction (Hypothesis):** Unclear PAN syntax rules (5 letters, 4 digits, 1 letter) and lack of plain-language explanation for why identity data is required trigger hesitation.

DIAGNOSIS · LEAK ZONE B (6,000 POST-KYC DROP-OFFS)

Why 66.67% of KYC-verified users do not invest

- **Dead-end verification state (Hypothesis):** After submitting KYC details, an unclear “pending” state leaves users unsure whether their steps succeeded or what comes next.
- **First-investment anxiety (Hypothesis):** Without consistent reinforcement that they can start with just **₹100**, newly verified beginners hesitate to commit capital.

EVIDENCE DISCIPLINE & EPISTEMIC HONESTY

Separating Supplied Facts, Methodological Limits & Design Hypotheses

Rigorous UX research requires being transparent about what the data actually proves—and what it cannot prove. The assignment context states that prior research consisted of a **survey of 40 existing users**, which **excluded users who dropped off**. This section documents the exact boundaries of that evidence and avoids inventing user quotes, interview findings, or survey percentages.

CRITICAL LIMITATION 01 · SURVIVORSHIP BIAS

40-User Survey Excluded All 37,000 Drop-Offs

Sampling only **40 existing users** means the survey captured feedback exclusively from the *surviving cohort* who successfully navigated onboarding, while excluding the **31,000 pre-KYC drop-offs (77.50%)** and **6,000 post-KYC drop-offs (66.67% of KYC completions)**.

Implication: Existing users can report what they liked about the app, but they cannot directly reveal the blocking friction (e.g., unlinked Aadhaar mobile numbers, PAN format confusion, or language barriers) that caused 92.50% of downloads to leave.

CRITICAL LIMITATION 02 · SAMPLE SIZE (N = 40)

0.10% of Total Downloads · 1.33% of Active Investors

A sample of `n = 40` represents just **0.10%** of the 40,000 downloads (`40 ÷ 40,000`) and **1.33%** of the 3,000 invested users (`40 ÷ 3,000`).

Implication: Any insights from a 40-person survivor sample must be treated as directional signals rather than statistically generalizable proof across India's diverse linguistic and socioeconomic beginner segments.

Evidence Classification Matrix: Facts vs. Decisions vs. Hypotheses vs. Future Experiments

Classification	Statement / Claim	Source or Validation Method
SUPPLIED FACT	Funnel counts: **40,000** downloads → **9,000** KYC completions (22.50%) → **3,000** first investments (7.50% of downloads, 33.33% of KYC).	Supplied directly in the Newtonite assignment brief.
SUPPLIED FACT	Prior survey covered **40 existing users** and excluded users who dropped off during onboarding or prior to investing.	Supplied directly in the Newtonite assignment brief.
DESIGN DECISION	Show the 3 required KYC items (PAN, Aadhaar, Aadhaar-linked mobile) on the Homepage and in an interactive Step 2 checklist before form entry.	Implemented in `src/components/Checklist.tsx` and `src/pages/onboarding/Step2Ready.tsx`.
DESIGN DECISION	Provide a 1-click **English / हिंदी** language toggle on Step 1 and lazy-load `Noto Sans Devanagari` when Hindi is selected.	Implemented in `LanguageSelector.tsx`, `i18n.ts`, and `OnboardingProvider.tsx`.
DESIGN DECISION	Separate Mobile Verification (Step 3: Phone → 6-box OTP → Verified) from KYC Preparation (Step 4: PAN regex check + sample filler + Aadhaar readiness).	Implemented in `Step3Mobile.tsx`, `OtpInput.tsx`, and `Step4Kyc.tsx`.
HYPOTHESIS H1	Front-loading the non-blocking document checklist in Step 2 will reduce mid-flow abandonment on Steps 3 & 4 by letting unprepared users pause deliberately and return with documents.	**Proposed Experiment:** Track Step 2 → Step 4 completion rate and return-session KYC completion rate via cohort analytics.
HYPOTHESIS H2	Explicit PAN format guidance (`ABCDE1234F`) and plain-language regulatory explanation on Step 4 will increase first-attempt PAN validation pass rates.	**Proposed Experiment:** Measure `kyc_pan_validation_error` rate per session before and after rollout.
HYPOTHESIS H3	Reinforcing the **₹100** starting step and summarizing completed steps on Step 5 will improve post-KYC conversion above the 33.33% baseline.	**Proposed Experiment:** Measure 14-day KYC-verified → First Investment conversion rate across cohorts.

Key Working Assumptions Made for the Prototype

- **Mobile-First Device Profile:** Most first-time Indian retail investors access onboarding on 360px–430px Android smartphones over variable mobile networks; therefore, tap targets (≥44px), single-column ergonomics, and lightweight assets (86 KB WebP mascot) take priority.
- **Prototype Safety Assumption:** Because this submission is an evaluative frontend prototype without a regulated backend, asking users to type real PAN or Aadhaar numbers would be irresponsible. All inputs are explicitly marked as demo-only (`DemoNotice`) and never leave browser memory.

PRODUCT STRATEGY

Turning a High-Friction KYC Funnel into a Calm, Guided Journey

Sprout 2.0 replaces the traditional "sign-up wall" with a transparent, educational progression. Every architectural choice maps back to reducing cognitive load, eliminating document surprises, and building trust without hype.

PRINCIPLE 01 · TRANSPARENCY BEFORE ACTION

Tell users what they need before asking them to type

Instead of dropping users straight into a PAN or Aadhaar input field, Sprout 2.0 previews the 3-step journey (*Get started → Complete verification → Explore investing*) and lists the 3 required documents on the homepage and again in Onboarding Step 2.

PRINCIPLE 02 · ONE COGNITIVE TASK PER SCREEN

Five focused steps with predictable navigation

Each onboarding screen has a single clear objective, a 5-segment progress bar (`Step X of 5`), a consistent **Back** / **Continue** action bar, and fits cleanly inside a `390 × 844` mobile viewport without vertical clutter.

PRINCIPLE 03 · BILINGUAL & JARGON-FREE COPY

Plain English and prototype Hindi (हिंदी)

Regulatory concepts are explained in everyday language (e.g., *"identity details are requested because financial rules require investors to be identified"*), with full onboarding translation toggleable on Screen 1.

PRINCIPLE 04 · RESPONSIBLE FINANCIAL DESIGN

Visible risk disclosures & zero return promises

No upward-shooting wealth charts, fake user testimonials, or return calculators promising fixed gains. Market risk is stated above the fold on the homepage, in the FAQ, and in the persistent footer of every onboarding screen.

End-to-End User Journey Architecture (Homepage → 5-Step Onboarding)

ENTRY · /
Homepage
₹100 value prop, above-fold risk note, 3 steps, readiness checklist & FAQs.

STEP 1 · `/onboarding/1`
Welcome
Mascot greeting + English / हिंदी language selector pill + Back to home.

STEP 2 · `/onboarding/2`
Get Ready
Interactive 3-item checklist (0 of 3 to 3 of 3 ready) for PAN, Aadhaar & phone.

STEP 3 · `/onboarding/3`
Mobile & OTP
Sub-states: 3A (+91 10-digit entry) → 3B (6-digit demo OTP) → 3C (Verified).

STEP 4 · `/onboarding/4`
KYC Details
Sample PAN regex check (`ABCDE1234F` helper) + Aadhaar readiness checkboxes.

STEP 5 · `/onboarding/5`
Pending
Mascot confirmation + dynamic *What you practised* summary + review/home actions.

How Information Architecture Decisions Map to Funnel Friction

Conventional Fintech Pattern	Why It Hurts First-Time Investors	Sprout 2.0 Redesigned Pattern
Immediate phone number prompt on first screen	Users bounce before understanding what the app offers or whether they need documents ready.	**Step 1 Welcome + Step 2 Get Ready** orient the user, offer language choice, and preview required documents first.
Raw PAN input with generic "Invalid input" error	Beginners do not memorize the 5-letter, 4-digit, 1-letter PAN structure and abandon after repeated errors.	**Explicit format hint (`ABCDE1234F`)**, auto-uppercase transformation, live format-ok badge, and **Use sample PAN** button.
Asking for full 12-digit Aadhaar number in early demo/prototype	Creates severe privacy anxiety and unnecessary sensitive-data exposure.	**Zero Aadhaar number collection:** Step 4 only asks users to confirm via checkboxes that their Aadhaar card is ready.
Opaque "Verification in progress" dead end	Leaves users uncertain whether their actions were recorded (66.67% post-KYC drop-off).	**Step 5 Summary Card (*What you practised*)** recaps checklist count, mobile demo status, and PAN format check.

DESIGN SYSTEM & INCLUSIVE CRAFT

Warm Cream & Forest Green Tokens, Typography & WCAG 2.1 AA Compliance

Defined in `src/styles/global.css` and `src/constants.ts`, Sprout 2.0's design system pairs a calm editorial palette with strict accessibility guarantees across contrast, touch targets, keyboard focus, and reduced motion.

1. Brand & Functional Colour Tokens (`src/styles/global.css`)

Forest Green	Mint Green	Light Green	Warm Cream	Charcoal	Error Red
`#0F5132`	`#E6F4EA`	`#CFEFD6`	`#FFF9F4`	`#1F2937`	`#B42318`
CTAs, brand, focus (8.9:1 on Cream)	Hero & checklist surfaces (8.1:1 w/ Forest)	Step badges, demo notices (7.1:1 w/ Forest)	Primary canvas & button text	Headings & body (14.1:1 on Cream)	Invalid border & alert (6.1:1 on Cream)

2. Typography Scale (Inter + Noto Sans Devanagari)

Role	Desktop / Mobile	Weight & Line Height
Hero H1	`60px` / `36px`	SemiBold 600 · 1.12 · -0.025em
Section H2	`36px` / `30px`	SemiBold 600 · 1.20 · -0.025em
Onboarding H1	`36px` / `28px`	SemiBold 600 · 1.25 · -0.025em
Card Title H3	`20px` / `20px`	SemiBold 600 · 1.40
Lead Body	`18px` / `18px`	Regular 400 · 1.625 (leading-relaxed)
Body / Buttons / Inputs	`16px` / `16px`	Regular 400 (Body) / SemiBold 600 (CTA)
Helper & Risk Note	`14px` / `14px`	Regular 400 · 1.625 (High readability)

3. Reusable Component Specifications

- **Primary Pill CTA (`ctaClass`):** `min-h-12` (`48px`), `px-7`, `rounded-full`, fill `#0F5132`, text `#FFF9F4`, hover `#0B3F27`, active `scale(0.98)`.
- **Secondary Pill Button (`secondaryBtnClass`):** `min-h-12` (`48px`), `rounded-full`, border `1px solid rgba(15,81,50,0.4)`, hover fill `#E6F4EA`.
- **Text / Phone / PAN Input (`inputClass`):** `min-h-12` (`48px`), `rounded-xl` (`12px`), fill `#FFFFFF`, border `rgba(15,81,50,0.3)`, invalid border `#B42318`.
- **6-Digit OTP Boxes (`OtpInput.tsx`):** 6 equal flex boxes, `h-14` (`56px` tall), `20px` SemiBold centered digit, supports full 6-digit clipboard paste, arrow keys, and Backspace navigation.
- **Interactive Checklist Row (`CheckRow`):** `min-h-14` (`56px`), `rounded-2xl`, transitions from `bg-white/70` to `bg-mint/70` with a `28×28px` `#0F5132` check badge.

4. Verified Accessibility & Inclusive Design Audit (`src/styles/global.css` & Components)

WCAG AA Contrast

All text pairings exceed `4.5:1`: Charcoal on Cream is `14.1:1`; Muted body (`charcoal/80`) is `9.7:1`; Forest on Mint is `8.1:1`; Error Red on Cream is `6.1:1`.

Touch Ergonomics

`@media (pointer: coarse)` enforces `min-height: 44px` globally. Primary buttons/inputs are `48px` tall; checklist rows and OTP boxes are `56px` tall.

Keyboard & ARIA

Visible `3px solid #0F5132` `:focus-visible` ring; programmatic focus moves to step `<h1>` on route change and to invalid fields on error; `aria-invalid` & `role="status"` wired.

Reduced Motion

`@media (prefers-reduced-motion: reduce)` disables smooth scroll and collapses all CSS transition and animation durations to `0.01ms`.

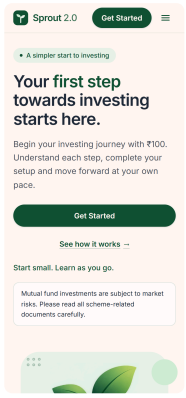
PUBLIC ACQUISITION EXPERIENCE (/)

Responsive Homepage Architecture & Visual Evidence

The homepage (`src/pages/Home.tsx`) adapts seamlessly from a 12-column editorial layout on Desktop (`1440px` , `max-w-6xl` container) to a single-column thumb-friendly flow on Mobile (`360px–430px`), keeping the **₹100 value proposition**, **Get Started CTA**, and **Market-Risk Disclaimer** visible above the fold on both viewports.



Desktop Full-Page Capture (1440 × 3080) — `design/screenshots/02-homepage-desktop-1440-full.png`



Mobile Above-the-Fold (390 × 844) — Hero, CTA & Risk Disclaimer

Above-the-Fold Discipline

On a standard `390 × 844` mobile screen, the visitor sees:

- Sticky `64px` Navbar with **Get Started** button & menu toggle
- Headline: *"Your first step towards investing starts here."*
- Supporting copy highlighting **₹100** starting amount
- Full-width `48px` **Get Started** CTA + *See how it works →*
- Bordered **Market-Risk Disclaimer** card before the fold

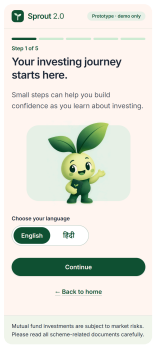
Section-by-Section Homepage Inventory (`src/components/*`)

Section & Component	Visual & Content Specification	Responsive Behavior (390px vs. 1440px)
1. Navbar `Navbar.tsx`	Custom geometric `SproutIcon` + `Sprout 2.0` wordmark, section links (*How It Works*, *Why Sprout*, *FAQs*), and primary **Get Started** pill.	Desktop shows inline pill links; Mobile collapses links into an accessible drawer with `Escape` close & scroll lock.
2. Hero Section `Hero.tsx`	Eyebrow pill, H1 headline, ₹100 intro, primary/secondary CTAs, above-the-fold risk card, and `HeroArt` with Sprout mascot + *"Your journey starts with ₹100"* card.	Desktop: 2-column asymmetric split (`1.05fr : 0.95fr`). Mobile: stacked single-column with text & risk card prioritized first.
3. How It Works `HowItWorks.tsx`	Mint section (`bg-mint/60`) with 3 cream cards: *Step 1 Get started*, *Step 2 Complete verification*, *Step 3 Explore investing*.	Desktop: 3-column equal-height card grid (`md:grid-cols-3`). Mobile: vertical card stack with `20px` gap.
4. Why Sprout `WhySprout.tsx`	4 numbered editorial rows: (1) *Start with ₹100*, (2) *Simple, beginner-friendly guidance*, (3) *Clear verification instructions*, (4) *Transparent about risk*.	Desktop: `0.8fr` sticky-style heading column + `1.2fr` numbered list. Mobile: stacked heading above list.
5. Getting Ready & FAQs `Checklist.tsx` , `Faq.tsx`	Rounded `#E6F4EA` readiness card (PAN, Aadhaar, linked mobile) + 4-item accordion (*Can I start with ₹100?* open by default) + `#0F5132` Footer.	Checklist transitions from 1-col mobile to 2-col desktop; FAQ stays centered at `max-w-3xl` (`768px`) for optimal line length.

INTERACTIVE ONBOARDING PROTOTYPE (/ONBOARDING/1 – /ONBOARDING/5)

Five-Screen Progressive Onboarding & Bilingual (हिंदी) Support

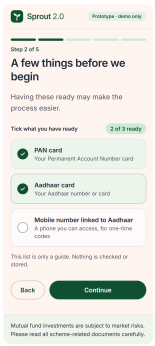
Every onboarding screen uses `OnboardingLayout.tsx` (`max-w-2xl` container, `Prototype · demo only` badge, 5-segment progress bar, and persistent bottom risk disclaimer). Below are all **8 captured mobile screen states (390 × 844)** from `design/screenshots/` :



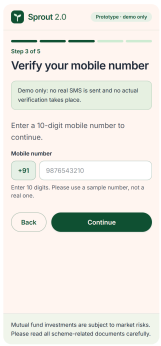
1A. Welcome (English)



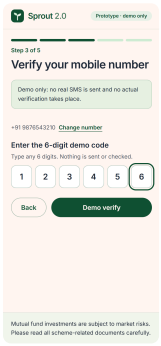
1B. Welcome (हिंदी Active)



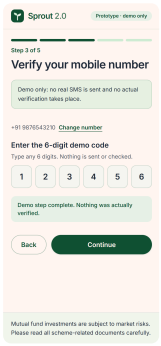
2. Get Ready (2 of 3 ready)



3A. Mobile Entry (+91)



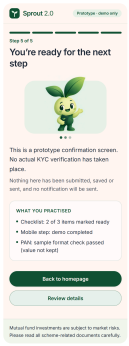
3B. 6-Digit Demo OTP



3C. Demo Verified State



4. KYC Details (ABCDE1234F)



5. Verification Pending

Detailed Specification of the 5 Onboarding Screens & Hindi (हिंदी) Localization

Screen	Interactive Controls & Validation Logic	Prototype Honesty & Bilingual (हिंदी) Behavior
Screen 1: Welcome `Step1Welcome.tsx`	Displays Sprout mascot, segmented `English` / `हिंदी` radio pills (`LanguageSelector.tsx`), full-width **Continue** CTA, and `← Back to home` link.	Selecting `हिंदी` switches `<html lang="hi">` , lazy-loads `Noto Sans Devanagari` , and shows the prototype translation notice: *"हिंदी अनुवाद प्रोटोटाइप है और अधूरा हो सकता है।"*
Screen 2: Get Ready `Step2Ready.tsx`	3 interactive `CheckRow` cards (*PAN card*, *Aadhaar card*, *Mobile number linked to Aadhaar*) with live counter pill (`0 of 3 ready` → `3 of 3 ready`).	Non-blocking guide so users can explore freely; explicitly states: *"This list is only a guide. Nothing is checked or stored."*
Screen 3: Mobile `Step3Mobile.tsx`	**3A:** `+91` badge + 10-digit numeric input (rejects non-digits/short input). **3B:** 6-box `OtpInput` with paste & arrow-key support + `Change number` link. **3C:** Read-only verified state with confirmation banner.	Top `DemoNotice` banner: *"Demo only: no real SMS is sent and no actual verification takes place."* Prompts user to enter a sample number, not a real one.
Screen 4: KYC Details `Step4Kyc.tsx`	Uppercase PAN input tested against `/^[A-Z]{5}[0-9]{4}[A-Z]$/` , one-tap **Use sample PAN** (`ABCDE1234F`), and 2 Aadhaar readiness checkboxes (first required).	PAN string lives strictly in local `useState('')` and is discarded on navigation. Never asks for an Aadhaar number.
Screen 5: Pending `Step5Pending.tsx`	Sprout mascot with 3 calm status dots, confirmation copy, dynamic **What you practised** summary card, **Back to homepage** CTA, and **Review details** button.	Clearly states: *"This is a prototype confirmation screen. No actual KYC verification has taken place."*

FRONTEND ENGINEERING & VERIFICATION

React 19 + Tailwind CSS 4 Implementation, Asset Budgets & Figma Handoff

Sprout 2.0 is built as a fast, zero-bloat single-page application in `C:/Users/A/OneDrive/Desktop/sprout-2.0`. Rather than pulling in heavy component or animation libraries, all interactions, routing, accessibility focus management, and bilingual translations are implemented in clean, auditable TypeScript.

BUILD & LINT VERIFICATION

0 Errors

`tsc -b && vite build + oxlint`

30 source files checked across 116 lint rules in 88ms; production bundle built in 706ms.

GZIPPED CODE FOOTPRINT

88.9 kB

Total JS + CSS + HTML (Gzipped)

JS: 81.60 kB gzip (266.80 kB raw) · CSS: 6.94 kB gzip (32.82 kB raw) · HTML: 0.39 kB gzip.

MASCOT & FONT OPTIMIZATION

86.2 kB

Lossless WebP Mascot (532 × 720)

Devanagari font (121.19 kB WOFF2) is lazy-loaded via dynamic `import()` only when हिंदी is selected.

1. Codebase Architecture (`src/`)

- **Custom Sync Router (`src/router.ts`):** 35-line `useSyncExternalStore` wrapper over the browser History API supporting `/` and `/onboarding/1` – `5` without external router overhead.
- **Privacy-Safe State (`src/onboarding/`):** Shared React context stores only non-sensitive session flags (`lang` , `ready` checklist booleans, demo `phone` , `phoneStage` , `panFormatChecked` , `aadhaarReady`).
- **Ephemeral PAN Handling (`Step4Kyc.tsx`):** As documented in source comments, the PAN value is held exclusively in component-local `useState('')` and disappears immediately upon leaving Screen 4.
- **Self-Hosted Variable Fonts:** Uses `@fontsource-variable/inter` and on-demand `@fontsource-variable/noto-sans-devanagari` so pages render identically offline or behind corporate firewalls without Google Fonts CDN requests.

2. UI/UX & Functional QA Matrix Verified

Test Area	Verification Scope & Result
Responsive Viewports	Inspected at `360px` , `390px` , `430px` , and `1440px` widths; zero horizontal scrollbar or clipped text.
Phone & OTP Flow	Tested empty/short phone errors, 6-digit paste (`123456`), ArrowLeft/ArrowRight, Backspace, and `Change number` reset.
PAN & KYC Validation	Tested lowercase auto-capitalization, invalid regex alert (`#B42318`), **Use sample PAN** (`ABCDE1234F`), and Aadhaar checkbox guard.
Bilingual Toggle	Verified instant switch between `English` and `हिंदी` across all 5 steps and back/review navigation.

3. Figma Handoff Kit & User-Provided Figma Links

To bridge code and design, the repository includes `docs/figma-handoff.md` , 14 standalone vector/raster assets in `design/figma-assets/` , 6 multi-layer editable SVG page boards in `design/figma-import-svgs/` , and 12 `@2x` reference screenshots in `design/screenshots/` .

Figma Artifact	Provided URL	Verification Note
Figma Design File (3 Free Starter Pages: Cover & Design System, Homepages, Onboarding EN/HI)	https://www.figma.com/design/67FVguQmfCpgOL2XGjKZge/Sprout-2.0-%E2%80%94-Newtonite-UI-UX-Assignment	User-provided link from Figma Share modal (verify *Anyone with link → can view* in incognito).
Figma Interactive Prototype (Connected Onboarding Flow 1 & Flow 2)	https://www.figma.com/proto/67FVguQmfCpgOL2XGjKZge/Sprout-2.0-%E2%80%94-Newtonite-UI-UX-Assignment?node-id=0-1&t=5Vwakxuj7i3lecnq-1	User-provided prototype link (`node-id=1-652`).

EXECUTION ROADMAP & MEASUREMENT

Three-Week Single-Developer Launch Plan, Risk Mitigation & Metrics

Below is a realistic **3-week engineering and rollout plan for a single frontend developer** transitioning Sprout 2.0 from prototype to production, followed by the exact telemetry definitions that would evaluate impact against the **22.50% KYC** and **33.33% post-KYC** baselines.

1. Three-Week Implementation & Rollout Timeline (1 Frontend Developer)

Timeline	Sprint Focus	Key Frontend & UX Deliverables	Exit Criteria
Week 1 *Days 1–5*	**Design Tokens & Responsive Homepage**	Configure Vite + Tailwind 4 tokens, focus/reduced-motion rules, custom SVG icons, optimized WebP mascot, and all 7 Homepage sections (`Navbar` to `Footer`).	Homepage passes WCAG AA contrast & 360px–1440px layout audit.
Week 2 *Days 6–10*	**5-Step Onboarding & Hindi Prototype**	Build `OnboardingLayout` , Steps 1–5, 6-box `OtpInput` , PAN regex validator, `i18n.ts` EN/HI dictionary, and lazy-loaded Devanagari font.	End-to-end 5-step flow functional in English & हिंदी with zero lint/type errors.
Week 3 *Days 11–15*	**API Integration, Telemetry & Staged Launch**	Wire privacy-safe funnel analytics hooks, integrate production SMS OTP & regulated KYC APIs behind feature flags, native Hindi copy review, and 10% → 100% rollout.	Production telemetry verified; zero PII leaked in client logs.

2. Proposed Success Metrics & Exact Measurement Methodology (No Fabricated Results)

Note: Because this project is a pre-launch design and prototype submission, no A/B test results or post-launch conversion lifts are fabricated. In production, success would be measured across 30-day cohorts using these exact definitions:

Metric Name	Exact Formula (Numerator ÷ Denominator)	Baseline Reference	How It Is Instrumented
1. Download / Visit → KYC Completion Rate	`Verified KYC users ÷ Users entering onboarding`	**22.50%** (`9,000 ÷ 40,000`)	Server `kyc_verified` event ÷ unique `onboarding_step_viewed(1)` cohort.
2. Step-to-Step Onboarding Drop-Off Rate	`Users completing Step N ÷ Users viewing Step N` (Steps 1–5)	**77.50%** cumulative pre-KYC loss	Client route events `step_1_viewed` … `step_5_viewed` segmented by checklist usage.
3. First-Attempt PAN Format Pass Rate	`Step 4 valid submits on 1st try ÷ Total Step 4 submit clicks`	New diagnostic metric (Tests Hypothesis H2)	Boolean event `pan_regex_passed_first_try` (never logs the PAN value itself).
4. Post-KYC First Investment Rate (14-Day)	`KYC users investing ≥ ₹100 within 14d ÷ Total KYC-verified users`	**33.33%** stage-to-stage (`3,000 ÷ 9,000`)	Order settlement webhook joined with `kyc_verified` user cohort.
5. End-to-End Funnel Conversion	`Users completing first investment ÷ Total app downloads`	**7.50%** end-to-end (`3,000 ÷ 40,000`)	Cumulative cohort ratio tracked weekly across English and `हिंदी` sessions.

3. Key Risks & Practical Mitigations

- **Prototype Hindi Copy Nuance:** Current Hindi strings are prototype translations. *Mitigation:* Specialist financial-translator review and comprehension testing before full release.
- **Live SMS & KYC Provider Latency:** Real telecom OTPs and KRA/CKYC lookups can time out. *Mitigation:* Add a 30s resend timer, clear error recovery states, and async status updates.
- **Unobserved Drop-Off Causes (Survivorship Bias):** Historical data only surveyed 40 existing users. *Mitigation:* Add an optional 1-question exit micro-poll for users leaving Steps 2–4.

CONCLUSION

Small Steps. Big Futures.

Sprout 2.0 addresses the core reasons first-time Indian investors drop out between download (40,000), KYC (9,000), and first investment (3,000). By pairing an approachable **₹100 starting point** with **upfront document readiness**, **step-by-step English & हिंदी guidance**, and **uncompromising risk transparency**, the experience replaces intimidation with clarity.

Mandatory Disclaimer: Mutual fund investments are subject to market risks. Please read all scheme-related documents carefully.